Alexander Soto

U.S. Citizen | +1(479) 306-2278 | alexsoto@stanford.edu | Project Portfolio | linkedin.com/in/alexsoto

EDUCATION

Stanford University — Stanford, CA
B.S. Electrical Engineering; Concentration: Hardware & Software; GPA: 3.88 — Sep. 2022 - June 2026

Stanford University — Stanford, CA
M.S. Electrical Engineering — Sep. 2026 - June 2027

PROFESSIONAL EXPERIENCE

Stanford University - Electrical Engineering Research Intern | View Poster — June 2025 – Present

- **Training deep learning models** to improve the speed and accuracy of PET (Positron Emission Tomography) image reconstruction.
- Generating simulated PET imaging data using the SimSET Monte Carlo toolkit.
- Enhancing and documenting the **Deep Learning Filtered Backprojection pipeline** for future researchers.
- Learning the fundamentals of PET imaging, including physics and image reconstruction.

Tyson Foods - Robotics/Machine Learning Intern | Springdale, AR | UMI Research — June 2024 – Sept. 2024

- Explored Universal Manipulation Interface (UMI), a system used to **transfer human skills to robots**, to further automate Tyson's plants.
- Assembled a pair of UMI grippers and collected robust training demonstrations.
- **Trained diffusion policies** modeling various human tasks.
- Learned the fundamentals of state-of-the-art **behavior cloning**.

University of Arkansas - Biomedical Engineering Research Intern | View Poster — June 2023 – Aug. 2023

- **Trained a convolutional neural network** in MATLAB to segment cells in NADH/FAD autofluorescence images to facilitate metabolic analysis.
- Fit a model that performed with **86% accuracy** on out of sample test images.
- Presented results at the national Biomedical Engineering Society conference.

PROJECTS

**Curling Robot | ** View Project

- Collaborated with a team to design and build a **fully autonomous robot** capable of delivering curling pucks into a central target ring.
- Designed and implemented the **IR receiver circuit**, incorporating a **photodetector, non-inverting amplifiers, filtering stages, and a comparator with hysteresis** to reliably detect infrared signals from target beacons.
- Enabled the robot to orient itself by rotating until the IR signal was detected.

**AC-DC Converter & Speaker | ** View Project

- Built an AC–DC power conversion circuit, implementing **full-wave rectification and voltage regulation** to produce a steady and adjustable DC output.
- Assembled a **functional speaker system** powered by the AC-DC converter.

**FPGA Music Player | ** View Project

- Built a real-time FPGA music player in **Verilog with chords, dynamics, and echo effects**.
- **Implemented an interactive UI** featuring effect-toggle buttons and a live waveform visualization.

**MIPS Processor | ** View Project

- **Implemented a 5-stage pipelined MIPS processor in Verilog**, incorporating full hazard detection, data forwarding, and branch handling.
- Validated the performance of the processor by **executing an optimized image convolution program** on the architecture.